

Extraneous Solutions and Real-World Applications

TEACHER KEY | Algebra II Honors | Unit 2 | Day 6 | TEK A2.6.D, A2.6.E

Objective: I can identify extraneous solutions - including cases where every solution is extraneous - and explain why checking in the original equation is necessary.

Vocabulary

Extraneous solution — a value the algebra produces that does NOT satisfy the ORIGINAL equation.

Part 1 — Why Checking the ORIGINAL Equation Matters

After splitting $|A| = k$ into $A = k$ or $A = -k$, both branches are true equations by construction. Explain why a solution can still fail once you plug it back into the ORIGINAL absolute value equation.

Sample: **the split only guarantees the two branches are consistent with EACH OTHER, not that the isolated expression was actually non-negative (or actually negative) the way that branch assumed - the check catches a branch used with the wrong sign**

Part 2 — A Solution That Does Not Check Out

Solve $3|x - 1| - 2 = 6x + 1$ and check BOTH answers.

$$|x - 1| = 2x + 1 \rightarrow x = -2 \text{ or } x = 0$$

Check $x = -2$: $3|-3| - 2 = 7$, $6(-2) + 1 = -11$. **EXTRANEIOUS** Check $x = 0$: $1 = 1$. **valid**

Part 3 — When EVERY Solution Is Extraneous

Solve $|x + 2| = -x - 5$ and check both.

$$\text{Split: } x + 2 = -x - 5 \text{ or } x + 2 = x + 5$$

$x = -3.5$ or the second equation gives **no solution** ($2 \neq 5$)

Check $x = -3.5$: $|-1.5| = 1.5$, $-(-3.5) - 5 = -1.5$. $1.5 \neq -1.5 \rightarrow$ **EXTRANEIOUS**

Solution set: **empty set - no valid solution at all**

Why did this happen? Look at the right side of the original equation for a clue.

Sample: **the right side, $-x - 5$, would have to be ≥ 0 for any x to work, since an absolute value can never equal a negative number - check whether that is even possible**

Part 4 — Finish the Other Two Day 4 Equations

Clarissa's houses: $|x - 1248| = 5 \rightarrow x = 1253$ or $x = 1243$ Both are even-numbered addresses within 5 houses, so **both are valid**

Surveying: $1.5 = |x - 10|/10 \cdot 100 \rightarrow |x - 10| = 0.15 \rightarrow x = 10.15$ or $x = 9.85$ miles

Part 5 — Finish Brighton Builders' Hallway

$$2|15x| + 52 = 292 \rightarrow |15x| = 120 \rightarrow x = 8 \text{ or } x = -8 \text{ (x is a suite count, so use } x = 8)$$

That is 8 office suites per side per floor $\rightarrow$ **32** suites per floor $\rightarrow$ **608** total suites in the 20-story building

Why was $x = -8$ rejected here, even though it is not technically extraneous the way Part 2's answer was?

a negative number of suites has no real-world meaning, even though it solves the equation algebraically

Part 6 — One More: Predict Before You Check

Solve $|2x + 5| = 3x + 1$. Before checking, predict: could BOTH solutions be valid, based on what the right side needs to be?

Prediction: **the right side $3x+1$ must be ≥ 0 for a solution to work, i.e. $x \geq -1/3$, so any candidate below that will fail**

Split: $2x + 5 = 3x + 1$ or $2x + 5 = -(3x + 1)$

$x = 4$ or $x = -1.2$

Check $x = 4$: $|13| = 13$, $3(4)+1 = 13$. **valid** Check $x = -1.2$: $|2.6| = 2.6$, $3(-1.2)+1 = -2.6$. **EXTRANEIOUS**
(matches the prediction - $-1.2 < -1/3$)

Practice

1. Solve $|2x - 1| = x + 4$ and check both. $x = 5$ or $x = -1$ (both valid)
2. Solve $|x - 6| = -x - 2$ and determine if any solution survives checking. **no valid solution**
3. Before solving $|x + 10| = 2x - 3$, what must be true of x for any solution to be valid? **$x \geq 1.5$**
4. Solve $|x + 10| = 2x - 3$ and check. $x = 13$ ($x = -7/3$ fails the $x \geq 1.5$ requirement, so it is rejected without even checking)
5. Solve $|4x - 1| = x + 8$ and check both. $x = 3$ or $x = -7/5$ (both valid, since $x + 8 \geq 0$ for both)
6. Solve $|5x + 2| = -3x - 5$. First state what x must satisfy for a solution to be possible, then solve, check, and give the solution set.

Requirement: **$x \leq -5/3$** $x = -7/8$ or $x = 1.5$ - neither satisfies $x \leq -5/3$, so both are extraneous. Solution set: **empty**

Exit Ticket

Solve $|3x + 1| = x - 7$. Check both, and determine the solution set. Explain your reasoning for rejecting anything you reject.

$x = -4$ and $x = 1.5$, but the right side $x-7$ must be ≥ 0 for any solution to be valid, and neither makes $x-7 \geq 0$ - no valid solution